

Title: MathSLM: A Transformer-Based Small Language Model for Mathematical Reasoning

Abstract

Mathematical reasoning is a fundamental capability required in educational systems, scientific computing, and intelligent tutoring applications. While large language models have demonstrated strong performance on mathematical problem solving, their high computational requirements and deployment costs make them unsuitable for many educational and resource-constrained environments. Existing rule-based mathematical solvers and traditional machine learning approaches are often limited to predefined problem types and fail to generalize across diverse mathematical reasoning tasks. This project proposes **MathSLM: A Transformer-Based Small Language Model for Mathematical Reasoning**, a lightweight Transformer-based framework designed to understand, reason, and generate step-by-step solutions for mathematical problems while maintaining computational efficiency.

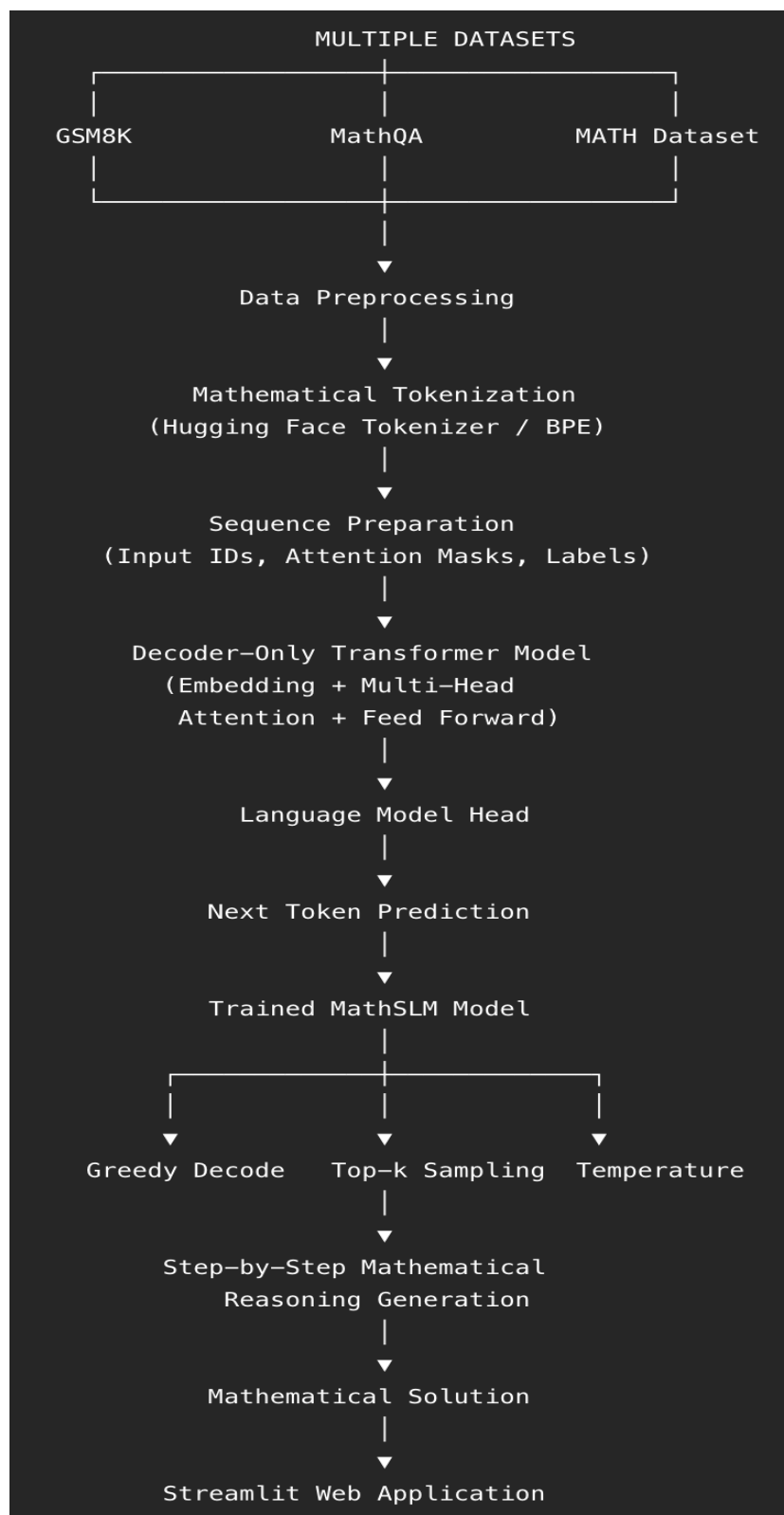
The proposed system implements a **decoder-only Transformer** architecture using the **PyTorch** deep learning framework and trains it on publicly available mathematical reasoning datasets. The framework performs data preprocessing, tokenization, sequence preparation, Transformer-based language modeling, supervised training, and autoregressive text generation. The model learns mathematical patterns through next-token prediction and is capable of generating intermediate reasoning steps before producing the final answer. The system further incorporates configurable text generation strategies such as greedy decoding, temperature sampling, and top-k sampling to improve response quality during inference. A web-based interface enables users to enter mathematical queries and interact with the trained model in real time.

The performance of the proposed **Transformer-based Small Language Model** is evaluated using training and validation loss, perplexity, token prediction accuracy, and benchmark evaluation on mathematical reasoning datasets. The generated responses are further assessed for correctness, coherence, and reasoning quality through qualitative analysis and sample-based evaluation. The resulting system aims to provide an efficient and accessible mathematical reasoning assistant that demonstrates the practical capabilities of Small Language Models while serving as a foundation for future research in domain-specific language models, mathematical intelligence, and educational AI applications.

Datasets

Dataset	URL Link
GSM8K (Grade School Math 8K)	https://huggingface.co/datasets/gsm8k
MathQA	https://huggingface.co/datasets/math_qa
MATH Dataset	https://huggingface.co/datasets/hendrycks/competition_math
NuminaMath (Optional)	https://huggingface.co/datasets/AI-MO/NuminaMath

Approach (Diagram):



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